

A pole-free torus one-point Virasoro-block recursion at $c = \frac{1}{2}$

Abstract

We construct a coefficientwise finite recursion for the eta-reduced torus one-point Virasoro block directly at the Ising central charge. Degenerate weights and their collision classes are classified exactly. At every grade, generic Kac recursion terms are grouped by their common Ising weight, and each fixed-central-charge Laurent coefficient is an exact collision moment: the constant term of a finite sum of Laurent germs. A valuation bound proves that only finitely many moments occur, and a grade-decreasing shifted recursion computes them. Rational continuation and partial-fraction uniqueness prove coefficientwise equality with the inverse-Shapovalov definition to all orders in q . Local Smith and crossing-form results identify the representation-theoretic meaning of the moments. The first collision gives a simple recursive residue at grade four; at grade ten its embedding diamond produces one double pole, whose complete principal part is evaluated explicitly. The quotient identity specialization recovers the three Ising characters from the corresponding BGG resolutions.

1 Introduction

For generic central charge, the eta-reduced torus one-point Virasoro block obeys the residue recursion derived in [2, Section 3, equations (7)–(9)]. At rational values of the Coulomb-gas parameter, distinct Kac labels have the same degenerate weight and the individual generic-recursion summands develop parameter poles. Javerzat, Santachiara, and Foda analyzed the cancellation of these non-physical poles in examples and in the identity-insertion character problem [3, Sections 4–6, especially equations (60)–(61)]. Stocco subsequently constructed central-charge-pole-free torus expressions through fourth order [4, Sections II.B–II.C and Appendix A]. Ribault identifies a recursion at rational β^2 as an open problem and singles out torus one-point Virasoro blocks as its simplest incarnation [5, Section 4.3.3].

This paper solves that problem at $c = 1/2$ for the meromorphic Verma-module block. “Pole-free” here means free of the divergent, non-physical parameter poles of the generic recursion; the genuine Laurent poles in the internal weight Δ remain. At every grade we partition the generic Kac summands into their exact Ising collision classes. Each class has a regular sum, and its fixed- c coefficients are finite constant terms of finite Laurent sums. The shifted recursion decreases the remaining grade, while an explicit valuation bound limits both the pole multiplicity and the number of Laurent coefficients required. Theorem 7.4 proves that this construction equals the inverse-Shapovalov definition at every power of q .

The local calculations exhibit how the algebraic confluence records Virasoro embedding geometry. The first collision is transverse and gives the level-four residue in Theorem 5.9. The first embedding diamond appears at level ten, creates a double pole, and is evaluated completely in Theorem 6.5. Finally, Theorem 8.3 distinguishes the Verma identity specialization from the irreducible quotient and recovers all three Ising characters.

2 Blocks and the reduced series

Definition 2.1 (Virasoro Verma module). The Virasoro algebra has generators L_n , $n \in \mathbb{Z}$, and central element c , with

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

The Verma module $M(c, \Delta)$ is generated by a vector $|\Delta\rangle$ such that

$$L_0|\Delta\rangle = \Delta|\Delta\rangle, \quad L_n|\Delta\rangle = 0 \quad (n > 0).$$

For a partition $Y = (y_1 \geq \dots \geq y_\ell)$, set $L_{-Y} = L_{-y_1} \cdots L_{-y_\ell}$, $|Y| = \sum_i y_i$, and $\ell(Y) = \ell$.

Definition 2.2 (Shapovalov and torus matrices). At level N , let

$$G_N(c, \Delta)_{Y,Y'} = \langle \Delta | L_Y L_{-Y'} | \Delta \rangle, \quad |Y| = |Y'| = N,$$

where $L_Y = L_{y_\ell} \cdots L_{y_1}$. For a primary field $V_d(z)$ normalized by $\langle \Delta | V_d(z) | \Delta \rangle = z^{-d}$, let

$$\rho_N(c, \Delta, d)_{Y,Y'} = \langle \Delta | L_Y V_d(1) L_{-Y'} | \Delta \rangle.$$

Whenever G_N is invertible, define

$$F_N(c, \Delta, d) = \sum_{|Y|=|Y'|=N} (G_N(c, \Delta)^{-1})_{Y,Y'} \rho_N(c, \Delta, d)_{Y,Y'}.$$

Definition 2.3 (Eta-reduced block). Write

$$\mathcal{F}_\Delta^{(c)}(d | q) = q^{\Delta - c/24} \sum_{N \geq 0} F_N(c, \Delta, d) q^N$$

and define

$$H_\Delta^{(c)}(d | q) = \prod_{m \geq 1} (1 - q^m) \sum_{N \geq 0} F_N(c, \Delta, d) q^N = \sum_{N \geq 0} H_N(c, \Delta, d) q^N.$$

Lemma 2.4 (Polynomial matrix entries). *For fixed N , every entry of $G_N(c, \Delta)$ belongs to $\mathbb{Q}[c, \Delta]$, and every entry of $\rho_N(c, \Delta, d)$ belongs to $\mathbb{Q}[c, \Delta, d]$.*

Proof. Repeated use of the Virasoro commutator moves every positive mode in L_Y to the right. Positive modes annihilate $|\Delta\rangle$, while each L_0 and the central element act by Δ and c , respectively. This gives the first assertion.

For the second assertion use, in addition,

$$[L_n, V_d(z)] = z^n (z \partial_z + (n+1)d) V_d(z).$$

Move positive modes through $V_d(z)$, then move the remaining negative modes through $V_d(z)$ toward the bra. The process terminates because each commutation removes one mode from one side of the insertion. The remaining derivatives act on z^{-d} ; after setting $z = 1$, their values are polynomials in d . The other commutators contribute only polynomials in c and Δ . $\square$

3 Ising resonance classes

Definition 3.1 (Ising Kac weights). For $r, s \in \mathbb{Z}_{>0}$, define

$$\Delta_{r,s} = \frac{(4r - 3s)^2 - 1}{48}.$$

Set

$$\lambda_n = \frac{n^2 - 1}{48} \quad (n \in \mathbb{Z}_{\geq 0}).$$

Proposition 3.2 (Exact collision classification). *For positive pairs (r, s) and (r', s') , $\Delta_{r,s} = \Delta_{r',s'}$ if and only if one of the following holds for some $\ell \in \mathbb{Z}$:*

$$(r', s') = (r + 3\ell, s + 4\ell),$$

or

$$(r', s') = (3\ell - r, 4\ell - s).$$

Proof. The equality of the two weights is equivalent to

$$4r' - 3s' = \varepsilon(4r - 3s)$$

for some $\varepsilon \in \{1, -1\}$. If $\varepsilon = 1$, then $4(r' - r) = 3(s' - s)$. Since $\gcd(3, 4) = 1$, there is an $\ell \in \mathbb{Z}$ with $(r' - r, s' - s) = (3\ell, 4\ell)$. If $\varepsilon = -1$, then $4(r' + r) = 3(s' + s)$, and the same coprimality argument gives $(r' + r, s' + s) = (3\ell, 4\ell)$. The converse follows by direct substitution. $\square$

Corollary 3.3 (Distinct weights and local finiteness). *The set of distinct positive-label degenerate weights at $c = 1/2$ is*

$$\{\lambda_n : n \in \mathbb{Z}_{\geq 0}\}.$$

Every collision class contains infinitely many positive pairs. For each fixed level N , only finitely many classes contain a pair with $rs \leq N$.

Proof. For every positive pair, $\Delta_{r,s} = \lambda_{|4r-3s|}$. Conversely, given $n \geq 0$, choose a sufficiently large positive $r \equiv n \pmod{3}$; then $s = (4r - n)/3$ is a positive integer and $4r - 3s = n$. Starting with any positive pair in a class, the pairs $(r + 3\ell, s + 4\ell)$, $\ell \geq 0$, are positive and remain in the same class. Finally, $rs \leq N$ implies $r \leq N$ and $s \leq N$, so only finitely many pairs, and hence classes, occur. $\square$

4 Coefficientwise partial fractions

Lemma 4.1 (Large internal weight). *For fixed c, d, N ,*

$$\lim_{\Delta \rightarrow \infty} F_N(c, \Delta, d) = p(N),$$

where $p(N)$ is the number of partitions of N . Consequently,

$$\lim_{\Delta \rightarrow \infty} H_N(c, \Delta, d) = \delta_{N,0}.$$

Proof. For a partition Y , put $a_Y = \ell(Y)$, and rescale the PBW vector $L_{-Y}|\Delta\rangle$ by $\Delta^{-a_Y/2}$. Commuting positive modes to the right shows that

$$\deg_{\Delta} G_N(Y, Y') \leq \min(a_Y, a_{Y'}).$$

If $a_Y = a_{Y'}$ and $Y \neq Y'$, the degree is at most $a_Y - 1$. For $Y = Y'$, the leading term is

$$\left(\prod_{j \geq 1} m_j(Y)! (2j)^{m_j(Y)} \right) \Delta^{a_Y},$$

where $m_j(Y)$ is the multiplicity of j in Y . Indeed, maximal degree requires every positive mode to be paired with a negative mode of the same index, using the $2jL_0$ term of $[L_j, L_{-j}]$; any different pairing uses at least one commutator without an L_0 contribution.

Apply the same commutation argument with one V_d insertion. The term in which no Ward commutator is used has the same leading term as the Shapovalov entry. Every term using $[L_n, V_d(z)]$ loses one possible positive–negative pairing. It follows that, after the above rescaling, both G_N and ρ_N converge to the same invertible diagonal matrix D_N . Therefore

$$\lim_{\Delta \rightarrow \infty} F_N(c, \Delta, d) = \text{tr}(D_N^{-1} D_N) = p(N).$$

If e_j are the coefficients of $\prod_{m \geq 1} (1 - q^m)$, then

$$H_N = \sum_{j=0}^N e_j F_{N-j}.$$

The generating-function identity

$$\left(\sum_{j \geq 0} e_j q^j \right) \left(\sum_{k \geq 0} p(k) q^k \right) = 1$$

gives the second limit. □

Theorem 4.2 (Finite coefficientwise pole expansion). *At $c = 1/2$, $H_0 = 1$, and for every $N \geq 1$ there are finite integers $m_{n,N}$ and coefficients $a_{n,k,N}(d) \in \mathbb{Q}[d]$ such that*

$$H_N \left(\frac{1}{2}, \Delta, d \right) = \sum_{\substack{n \geq 0 \\ \exists r,s > 0: rs \leq N, |4r-3s|=n}} \sum_{k=1}^{m_{n,N}} \frac{a_{n,k,N}(d)}{(\Delta - \lambda_n)^k}.$$

In particular, setting

$$A_{n,k}(d; q) = \sum_{N \geq 1} a_{n,k,N}(d) q^N$$

gives a coefficientwise locally finite expansion

$$H_{\Delta}^{(1/2)}(d | q) = 1 + \sum_{n \geq 0} \sum_{k \geq 1} \frac{A_{n,k}(d; q)}{(\Delta - \lambda_n)^k}.$$

Proof. By Lemma 2.4, $F_N = \text{tr}(G_N^{-1} \rho_N^{\top})$ is rational in Δ , and its poles are contained in the zero locus of $\det G_N$. The Kac determinant formula [1, Theorem 4.2 and equations (4.16)–(4.21)] gives, up to a nonzero factor independent of Δ ,

$$\det G_N(c, \Delta) = C_N \prod_{\substack{r,s > 0 \\ rs \leq N}} (\Delta - \Delta_{r,s}(c))^{p(N-rs)}.$$

At $c = 1/2$, Proposition 3.2 identifies the distinct zeros with the stated finite set of λ_n 's. Since H_N is a finite integer linear combination of $F_0, \dots, F_N$, it has the same pole containment. Lemma 4.1 shows that H_N tends to zero at infinity for $N \geq 1$, so its partial-fraction expansion has no polynomial part. All matrix entries have rational coefficients at $c = 1/2$, and their dependence on d is polynomial; hence the partial-fraction coefficients belong to $\mathbb{Q}[d]$. Local finiteness follows from Corollary 3.3. □

5 Local Smith exponents and the first collision

Lemma 5.1 (Pole order from local invariant factors). *Let $R = \mathbb{C}[[x]]$, and let $G(x) \in M_m(R)$ have nonzero determinant. Suppose the Smith exponents of G are $0 \leq e_1 \leq \dots \leq e_m$. Then*

$$\text{ord}_x \det G = \sum_{i=1}^m e_i, \quad \dim \ker G(0) = \#\{i : e_i > 0\},$$

and the pole order of $G(x)^{-1}$ is e_m . In particular, if

$$\text{ord}_x \det G = \dim \ker G(0) = k,$$

then every positive Smith exponent equals 1, and G^{-1} has at most a simple pole.

Proof. Since R is a discrete valuation ring, there are $U, V \in \text{GL}_m(R)$ such that

$$UGV = \text{diag}(x^{e_1}, \dots, x^{e_m}).$$

Taking determinants gives the first identity. Reducing modulo x gives the second. Inverting the displayed Smith form gives pole order e_m . If the sum of the k positive integers e_i is k , then each of them is 1. $\square$

Proposition 5.2 (Transverse inverse residue). *Let V be finite dimensional, let $G(x): V \rightarrow V^*$ be a symmetric matrix over $\mathbb{C}[[x]]$ with nonzero determinant, and put $K = \ker G(0)$. Let $\iota: K \hookrightarrow V$ be the inclusion and define the crossing form*

$$\beta(u, v) = u^\top G'(0)v, \quad u, v \in K.$$

If β is nondegenerate, then $G(x)^{-1}$ has only a simple pole and

$$\text{Res}_{x=0} G(x)^{-1} = \iota \beta^{-1} \iota^*.$$

Moreover, β is nondegenerate if and only if

$$\text{ord}_x \det G = \dim K.$$

Proof. Choose a complement C of K . Since K is the radical of the symmetric form $G(0)$, its restriction A to C is nondegenerate. In the decomposition $V = K \oplus C$, one therefore has

$$G(x) = \begin{pmatrix} xB + O(x^2) & xD + O(x^2) \\ xD^\top + O(x^2) & A + O(x) \end{pmatrix},$$

where B is the matrix of β . The Schur complement of the lower-right block is

$$S(x) = xB + O(x^2).$$

If B is invertible, block inversion gives

$$G(x)^{-1} = \frac{1}{x} \begin{pmatrix} B^{-1} & 0 \\ 0 & 0 \end{pmatrix} + O(1),$$

which is the coordinate expression of the asserted residue. Also

$$\det G(x) = \det(A + O(x)) \det S(x).$$

Every entry of S is divisible by x , so its determinant is divisible by $x^{\dim K}$, with equality of orders precisely when $\det B \neq 0$. This proves the final equivalence. $\square$

Proposition 5.3 (Two-step inverse principal part). *Let*

$$G(x) = G_0 + xG_1 + x^2G_2 + O(x^3)$$

be as in Proposition 5.2, and put

$$K_1 = \ker G_0, \quad \beta_1(u, v) = u^\top G_1 v, \quad K_2 = \ker(\beta_1: K_1 \rightarrow K_1^*).$$

For $v \in K_2$, choose $v_1 \in V$ satisfying

$$G_0 v_1 = -G_1 v$$

and define

$$\beta_2(u, v) = u^\top (G_2 v + G_1 v_1), \quad u, v \in K_2.$$

Then v_1 exists, β_2 is independent of its choice, and β_2 is symmetric. If β_2 is nondegenerate, then all Smith exponents are at most 2 and

$$[x^{-2}]G(x)^{-1} = \iota_2 \beta_2^{-1} \iota_2^*,$$

where $\iota_2: K_2 \hookrightarrow V$. In this case,

$$\#\{i : e_i \geq 1\} = \dim K_1, \quad \#\{i : e_i \geq 2\} = \dim K_2.$$

Proof. Because G_0 is symmetric, $\text{im } G_0$ is the annihilator of K_1 . For $v \in K_2$ and $u \in K_1$,

$$u^\top G_1 v = \beta_1(u, v) = 0.$$

Thus $G_1 v \in \text{im } G_0$, proving the existence of v_1 . Two choices differ by an element $k \in K_1$, and $u^\top G_1 k = \beta_1(u, k) = 0$ for $u \in K_2$, proving choice-independence.

Choose a complement to K_1 as in the proof of Proposition 5.2, and first take the Schur complement of its nondegenerate block. This produces a symmetric matrix on K_1 of the form

$$S(x) = xB_1 + x^2B_2 + O(x^3),$$

where B_1 represents β_1 , and the restriction of B_2 to $\ker B_1 = K_2$ is the displayed form β_2 . This also proves its symmetry. Now choose a complement E to K_2 in K_1 . The restriction of B_1 to E is nondegenerate, and a second Schur complement gives

$$S(x) \sim \begin{pmatrix} x^2\beta_2 + O(x^3) & 0 \\ 0 & xB_1|_E + O(x^2) \end{pmatrix}.$$

If β_2 is nondegenerate, block inversion yields the asserted x^{-2} coefficient. The same block form gives the two counts of Smith exponents. $\square$

Corollary 5.4 (Determinant multiplicity does not determine block pole order). *At a collided Kac weight, the multiplicity of the corresponding factor in $\det G_N$ is only the sum of the positive local Smith exponents. The inverse Shapovalov pole order is their maximum.*

Proof. Apply Lemma 5.1 with $x = \Delta - \lambda_n$. $\square$

Lemma 5.5 (Torus form on a singular submodule). *Let $\chi \in M(c, h)$ be a singular vector of level ℓ , normalized by a fixed PBW coefficient, and let*

$$P_\chi(d) = \langle \chi | V_d(1) | \chi \rangle.$$

For partitions Y, Y' of the same integer m ,

$$\langle \chi | L_Y V_d(1) L_{-Y'} | \chi \rangle = P_\chi(d) \rho_m(c, h + \ell, d)_{Y, Y'}.$$

Proof. The assignment

$$L_{-Y}|h + \ell\rangle \mapsto L_{-Y}\chi$$

is a homomorphism from $M(c, h + \ell)$ to $M(c, h)$, because χ is singular and has L_0 -weight $h + \ell$. Pull the torus three-point form on $M(c, h)$ back along this homomorphism in both module arguments. Commuting one Virasoro mode at a time through $V_d(z)$ shows that the pulled-back form obeys the same Ward recursion as $\rho_m(c, h + \ell, d)$. That recursion terminates at the highest-weight matrix element and therefore determines every descendant matrix element uniquely. The pulled-back highest-weight value is $P_\chi(d)$, whereas the normalization $\langle h + \ell | V_d(1) | h + \ell \rangle = 1$ gives highest-weight value 1 for ρ_m . The displayed identity follows. $\square$

Lemma 5.6 (First crossing form on a singular submodule). *Fix c and let $\chi = P|h_0\rangle \in M(c, h_0)$ be singular of level ℓ , where the PBW operator $P \in U(\text{Vir}_-)$ is held fixed as h varies. Set*

$$b_\chi = \left. \frac{d}{dh} \langle h | \omega(P) P | h \rangle \right|_{h=h_0},$$

where $\omega(L_n) = L_{-n}$. For partitions Y, Y' of m ,

$$\left. \frac{d}{dh} \langle h | \omega(P) L_Y L_{-Y'} P | h \rangle \right|_{h=h_0} = b_\chi G_m(c, h_0 + \ell)_{Y, Y'}.$$

Proof. The derivative on the left defines a bilinear form C on the PBW space of $M(c, h_0 + \ell)$. At $h = h_0$, the map

$$j_\chi: M(c, h_0 + \ell) \longrightarrow M(c, h_0), \quad u|h_0 + \ell\rangle \mapsto u\chi,$$

is a module homomorphism, and its image lies in the radical of the Shapovalov form.

Negative modes intertwine with j_χ identically. When a positive mode is commuted through $P|h\rangle$, the difference from the intertwining identity vanishes at h_0 and is therefore divisible by $h - h_0$. Its first-order contribution to C pairs with $\text{im } j_\chi$ in the specialized Shapovalov form and hence vanishes. Thus C is contravariant. A contravariant form on a Verma module is uniquely determined by its highest-weight value, because PBW commutation recursively reduces every matrix element to that value. Here the highest-weight value is b_χ , so $C = b_\chi G_m(c, h_0 + \ell)$. $\square$

Proposition 5.7 (First Ising collision). *At level 4, the labels $(1, 2)$ and $(2, 2)$ collide at $\Delta = 1/16$. In the PBW basis*

$$(L_{-4}, L_{-3}L_{-1}, L_{-2}^2, L_{-2}L_{-1}^2, L_{-1}^4)|\Delta\rangle,$$

the Shapovalov determinant at $c = 1/2$ is

$$12\Delta^3(2\Delta - 7)(2\Delta - 1)^3(3\Delta - 5)(16\Delta - 21)(16\Delta - 1)^3.$$

At $\Delta = 1/16$, the local Smith exponents are

$$(0, 0, 1, 1, 1).$$

Consequently, $G_4(1/2, \Delta)^{-1}$, $F_4(1/2, \Delta, d)$, and $H_4(1/2, \Delta, d)$ have at most a simple pole at $\Delta = 1/16$.

Proof. Direct Virasoro commutation gives

$$G_4 = \begin{pmatrix} 8\Delta + \frac{5}{2} & 14\Delta & 24\Delta + \frac{3}{2} & 36\Delta & 120\Delta \\ 14\Delta & 12\Delta^2 + 14\Delta & 30\Delta & 40\Delta^2 + 20\Delta & 192\Delta^2 + 48\Delta \\ 24\Delta + \frac{3}{2} & 30\Delta & 32\Delta^2 + 36\Delta + \frac{17}{8} & 48\Delta^2 + 51\Delta & 216\Delta^2 + 144\Delta \\ 36\Delta & 40\Delta^2 + 20\Delta & 48\Delta^2 + 51\Delta & 32\Delta^3 + 118\Delta^2 + 33\Delta & 288\Delta^3 + 336\Delta^2 + 96\Delta \\ 120\Delta & 192\Delta^2 + 48\Delta & 216\Delta^2 + 144\Delta & 288\Delta^3 + 336\Delta^2 + 96\Delta & 384\Delta^4 + 1152\Delta^3 + 1056\Delta^2 + 288\Delta \end{pmatrix}$$

Its determinant is the displayed polynomial. At $\Delta = 1/16$, the upper-left 2×2 minor has determinant 2, while the following three independent vectors lie in the kernel:

$$\begin{pmatrix} -9/16 \\ -3/2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -27/64 \\ -9/8 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -465/256 \\ -75/32 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Thus the specialized matrix has rank 2 and nullity 3. The determinant has order 3 at $1/16$, so Lemma 5.1 gives the stated Smith exponents and inverse pole bound. The definitions of F_4 and H_4 then give the same bound for the block coefficients. $\square$

Proposition 5.8 (Singular data at the first collision). *Put $h_0 = 1/16$. The following vectors in $M(1/2, h_0)$ are singular:*

$$\chi_2 = \left(L_{-1}^2 - \frac{3}{4}L_{-2} \right) |h_0\rangle$$

and

$$\chi_4 = \left(L_{-1}^4 - \frac{25}{6}L_{-2}L_{-1}^2 + \frac{49}{144}L_{-2}^2 + \frac{11}{6}L_{-3}L_{-1} - \frac{1}{4}L_{-4} \right) |h_0\rangle.$$

At level 4,

$$\ker G_4(1/2, h_0) = \text{span}\{L_{-2}\chi_2, L_{-1}^2\chi_2, \chi_4\}.$$

In this ordered basis the crossing form of Proposition 5.2, with $x = \Delta - h_0$, is

$$\beta_4 = \begin{pmatrix} -119/8 & -693/32 & 0 \\ -693/32 & -9471/128 & 0 \\ 0 & 0 & 13475/648 \end{pmatrix} = \begin{pmatrix} -\frac{7}{4}G_2(1/2, 33/16) & 0 \\ 0 & 13475/648 \end{pmatrix}.$$

Finally, the two highest-weight torus matrix elements are

$$P_2(d) := \langle \chi_2 | V_d(1) | \chi_2 \rangle = \frac{d(d-1)(2d-7)(2d-1)}{4}$$

and

$$\begin{aligned} P_4(d) &:= \langle \chi_4 | V_d(1) | \chi_4 \rangle \\ &= \frac{d(d-1)(2d-7)(2d-1)(3d-14)(3d-5)(6d-55)(6d-1)}{1296}. \end{aligned}$$

Proof. Apply $L_1, \dots, L_\ell$ to the displayed level- ℓ vector and use the Virasoro commutator. Collecting coefficients in the PBW bases at levels $\ell - 1, \dots, 0$ gives zero for χ_2 and for χ_4 ; modes larger than ℓ vanish by degree. Thus both vectors are singular.

In the level-four PBW basis of Proposition 5.7, the three asserted kernel vectors are the columns

$$\begin{pmatrix} 0 \\ 0 \\ -3/4 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -3/2 \\ -3/2 \\ 0 \\ -3/4 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} -1/4 \\ 11/6 \\ 49/144 \\ -25/6 \\ 1 \end{pmatrix}.$$

Multiplication by the displayed matrix $G_4(1/2, h_0)$ gives zero. They are independent, and Proposition 5.7 gives nullity three, proving the kernel identity.

Multiplying these columns on both sides of $\partial_\Delta G_4(1/2, \Delta)|_{\Delta=h_0}$ gives the displayed crossing matrix. Direct level-two commutation gives

$$G_2(1/2, 33/16) = \begin{pmatrix} 17/2 & 99/8 \\ 99/8 & 1353/32 \end{pmatrix},$$

which verifies the second expression for β_4 . Finally, repeated use of the primary-field Ward commutator in the proof of Lemma 2.4, followed by collection and factorization over $\mathbb{Q}[d]$, gives the displayed P_2 and P_4 . $\square$

Theorem 5.9 (First fixed- c recursive principal part). *At $c = 1/2$, the level-four reduced coefficient satisfies the finite fixed-central-charge recursion*

$$\text{Res}_{\Delta=1/16} H_4(1/2, \Delta, d) = -\frac{4}{7}P_2(d)H_2(1/2, 33/16, d) + \frac{648}{13475}P_4(d).$$

Equivalently, this residue is

$$\frac{d(d-1)(2d-7)(2d-1)}{40425} (248d^4 - 5752d^3 + 25274d^2 - 33630d + 5775).$$

Every quantity on the right is evaluated directly at $c = 1/2$.

Proof. At levels 2, 3, and 4, the radicals at $\Delta = h_0$ are respectively

$$\text{span}\{\chi_2\}, \quad \text{span}\{L_{-1}\chi_2\}, \quad \text{span}\{L_{-2}\chi_2, L_{-1}^2\chi_2, \chi_4\}.$$

The crossing forms on the first two spaces are

$$-\frac{7}{4}, \quad -\frac{7}{4}G_1(1/2, 33/16),$$

and the third is the form in Proposition 5.8. These identities follow by differentiating the level-two, level-three, and level-four Gram matrices; all three crossing forms are nondegenerate. Apply Proposition 5.2, then use Lemma 5.5 on each singular summand. Taking the trace against the corresponding torus matrix gives

$$\begin{aligned} \text{Res}_{\Delta=h_0} F_2 &= -\frac{4}{7}P_2, \\ \text{Res}_{\Delta=h_0} F_3 &= -\frac{4}{7}P_2F_1(1/2, 33/16, d), \\ \text{Res}_{\Delta=h_0} F_4 &= -\frac{4}{7}P_2F_2(1/2, 33/16, d) + \frac{648}{13475}P_4. \end{aligned}$$

The Euler product has coefficients 1, -1, -1, 0, 0 through degree four. Hence $H_4 = F_4 - F_3 - F_2$, while $H_2 = F_2 - F_1 - F_0$. Substitution gives the recursive formula. Inserting the explicit polynomials from Proposition 5.8 and the level-two coefficient from the inverse Shapovalov matrix gives the second displayed expression. $\square$

6 The first nontransverse embedding diamond

Definition 6.1 (Generic residue data used for algebraic confluence). For $b \in \mathbb{C}^\times$, set

$$c(b) = 1 + 6(b + b^{-1})^2, \quad h_{r,s}(b) = \frac{(b + b^{-1})^2 - (rb + sb^{-1})^2}{4}.$$

Put $\Lambda^2 = (b + b^{-1})^2 - 4d$, and define

$$\begin{aligned} \mathcal{P}_{r,s}(b, d) &= \prod_{\substack{1 \leq k \leq 2r-1 \\ k \text{ odd}}} \prod_{\substack{1 \leq l \leq 2s-1 \\ l \text{ odd}}} \frac{\Lambda^2 - (kb + lb^{-1})^2}{4} \\ &\quad \times \frac{\Lambda^2 - (kb - lb^{-1})^2}{4}, \end{aligned}$$

and

$$\mathcal{A}_{r,s}(b) = \frac{1}{2} \prod_{\substack{1-r \leq p \leq r, \ 1-s \leq q \leq s \\ (p,q) \neq (0,0), (r,s)}} (pb + qb^{-1})^{-1}, \quad \mathcal{R}_{r,s}(b, d) = \mathcal{A}_{r,s}(b) \mathcal{P}_{r,s}(b, d).$$

For generic b , $\mathcal{R}_{r,s}$ is the residue coefficient in the eta-reduced torus recursion.

Proof. The formula is equations (7)–(9) of [2, Section 3]; the product $\mathcal{P}_{r,s}$ is their product of the two torus fusion polynomials, written without choosing a square root of Λ^2 . $\square$

Proposition 6.2 (Grade-ten Smith profile at $\Delta = 1/16$). *Let $h_0 = 1/16$, and let*

$$U_2 = U(\text{Vir}_-)\chi_2, \quad U_4 = U(\text{Vir}_-)\chi_4$$

be the two singular submodules of $M(1/2, h_0)$ from Proposition 5.8. Their first intersection is generated at total level 10 by a singular vector χ_{10} , and

$$\dim \ker G_{10}(1/2, h_0) = 32.$$

The 42 local Smith exponents of $G_{10}(1/2, h_0 + x)$ are

$$(0^{10}, 1^{31}, 2).$$

In particular, the inverse Shapovalov matrix has a nonzero rank-one coefficient at order x^{-2} .

Proof. The weight h_0 is the interior Ising weight labelled by $(1, 2)$, so it is in Class R^+ , Case 1+, in the classification of [1, Section 5.1.4]. The commutative embedding diagram in [1, Proposition 5.4 and Figure 5.4] describes the two branches and their successive intersections. The identification of the first Jantzen layer with their sum, hence with the maximal proper submodule, is [1, Theorem 6.3(i) and equation (6.3)].

The two highest weights are

$$h_0 + 2 = \frac{33}{16} = \lambda_{10}, \quad h_0 + 4 = \frac{65}{16} = \lambda_{14}.$$

The first singular vector inside U_2 has relative level 8, with Kac label $(4, 2)$, and the first singular vector inside U_4 has relative level 6, with Kac label $(1, 6)$. Both therefore have total level 10 and weight

$$h_0 + 10 = \frac{161}{16} = \lambda_{22}.$$

The embeddings are injective by [1, the paragraph preceding Corollary 5.2]. Their two images are proportional by uniqueness of a singular vector at a fixed level, [1, Proposition 5.1]. Hence the level-ten intersection is one dimensional. There is no earlier intersection: a vector at the first intersection grade would be singular, whereas the Kac determinant for $M(1/2, 33/16)$, respectively $M(1/2, 65/16)$, has no zero below relative level 8, respectively 6.

The grade-ten radical is the grade-ten part of $U_2 + U_4$. Since

$$p(8) = 22, \quad p(6) = 11,$$

and the intersection has dimension one at this grade,

$$\dim \ker G_{10}(1/2, h_0) = 22 + 11 - 1 = 32.$$

By Proposition 3.2, the only positive Kac labels with $\Delta_{r,s} = h_0$ and $rs \leq 10$ are $(1, 2)$ and $(2, 2)$. The Kac determinant formula therefore gives

$$\text{ord}_x \det G_{10}(1/2, h_0 + x) = p(8) + p(6) = 33.$$

There are $p(10) = 42$ Smith exponents, exactly 32 of them are positive, and their sum is 33. Thus 31 positive exponents equal 1, one equals 2, and the remaining 10 vanish. Lemma 5.1 gives the final assertion. $\square$

Lemma 6.3 (Exact constants of the grade-ten confluence). *Let $b_0 = 2i/\sqrt{3}$, so $c(b_0) = 1/2$. Then*

$$\frac{(r, s)}{\mathcal{A}_{r,s}(b_0)} \left| \begin{array}{cccc} (1, 2) & (2, 2) & (4, 2) & (1, 6) \\ -4/7 & 648/13475 & 48/32035128125 & -8/320945625 \end{array} \right.$$

If a dot denotes d/db at b_0 , then

$$\frac{\dot{h}_{1,2} - \dot{h}_{2,2}}{\dot{h}_{1,2} - \dot{h}_{4,2}} = \frac{1}{5}, \quad \frac{\dot{h}_{2,2} - \dot{h}_{1,2}}{\dot{h}_{2,2} - \dot{h}_{1,6}} = \frac{1}{7},$$

and

$$\frac{1}{5} \mathcal{A}_{1,2} \mathcal{A}_{4,2} = \frac{1}{7} \mathcal{A}_{2,2} \mathcal{A}_{1,6} = -\frac{192}{1121229484375}.$$

Moreover,

$$\begin{aligned} \mathcal{P}_{4,2}(b_0, d) &= \frac{1}{20736} d(d-20)(d-13)(d-11)(d-6)(d-1) \\ &\quad \times (2d-57)(2d-35)(2d-15)(2d-7)(2d-5)(2d-1) \\ &\quad \times (3d-14)(3d-5)(6d-55)(6d-1), \end{aligned}$$

$$\begin{aligned} \mathcal{P}_{1,6}(b_0, d) &= \frac{1}{64} d(d-20)(d-13)(d-11)(d-6)(d-1) \\ &\quad \times (2d-57)(2d-35)(2d-15)(2d-7)(2d-5)(2d-1), \end{aligned}$$

and

$$P_2(d) \mathcal{P}_{4,2}(b_0, d) = P_4(d) \mathcal{P}_{1,6}(b_0, d).$$

Proof. Substitute $b_0 = 2i/\sqrt{3}$ in the finite products of Definition 6.1. Direct differentiation gives

$$c(b_0) = \frac{1}{2}, \quad c'(b_0) = \frac{7i\sqrt{3}}{2} \neq 0,$$

so $b - b_0$ is a valid local deformation parameter. The Kac-weight derivatives are

$$\frac{(r, s)}{16\dot{h}_{r,s}/(i\sqrt{3})} \left| \begin{array}{cccc} (1, 2) & (2, 2) & (4, 2) & (1, 6) \\ 9 & -7 & -71 & 105 \end{array} \right.$$

These four values give the two slope ratios. Multiplication of the four displayed $\mathcal{A}$ -values gives the common nested residue. Factoring the two $\mathcal{P}$ -products over $\mathbb{Q}[d]$ gives the last three identities. $\square$

Theorem 6.4 (Leading double-pole coefficient at grade ten). *Normalize all singular vectors by setting their L_{-1}^ℓ coefficient equal to 1. Let $\psi_8 \in M(1/2, 33/16)$ and $\psi_6 \in M(1/2, 65/16)$ be the singular vectors of levels 8 and 6, respectively. Then*

$$\chi_{10} = j_{\chi_2}(\psi_8) = j_{\chi_4}(\psi_6).$$

Define

$$P_8^{(33/16)}(d) = \langle \psi_8 | V_d(1) | \psi_8 \rangle, \quad P_6^{(65/16)}(d) = \langle \psi_6 | V_d(1) | \psi_6 \rangle.$$

They are the explicit polynomials

$$P_8^{(33/16)}(d) = \mathcal{P}_{4,2}(b_0, d), \quad P_6^{(65/16)}(d) = \mathcal{P}_{1,6}(b_0, d)$$

listed in Lemma 6.3. Write

$$G_{10}(1/2, h_0 + x) = G_0 + xG_1 + x^2G_2 + O(x^3),$$

choose v_1 with

$$G_0 v_1 = -G_1 \chi_{10},$$

and set

$$\gamma_{10} = \chi_{10}^T (G_2 \chi_{10} + G_1 v_1).$$

Then it is independent of the choice of v_1 , and

$$\gamma_{10} = -\frac{1121229484375}{192}.$$

Consequently,

$$\lim_{\Delta \rightarrow h_0} (\Delta - h_0)^2 H_{10}(1/2, \Delta, d) = \frac{P_2(d) P_8^{(33/16)}(d)}{\gamma_{10}} = \frac{P_4(d) P_6^{(65/16)}(d)}{\gamma_{10}}.$$

This is an intrinsic fixed- c expression for the first nontransverse Laurent coefficient.

Proof. The two compositions in the first display are nonzero singular vectors of total level 10. Proposition 5.1 of [1] makes them proportional. Their L_{-1}^{10} coefficients are both 1, so they are equal.

Let $K_1 = \ker G_0 = (U_2 + U_4)_{10}$, and let β_1 be its first crossing form. By Lemma 5.6, the restriction of β_1 to U_2 is the scalar b_{χ_2} times the Shapovalov form on $M(1/2, 33/16)$. Since ψ_8 is singular, $\beta_1(\chi_{10}, U_2) = 0$. The same argument through U_4 gives $\beta_1(\chi_{10}, U_4) = 0$. Therefore $\chi_{10} \in K_2 = \ker \beta_1$.

Proposition 6.2 shows that exactly one Smith exponent is at least 2. Hence K_2 is spanned by χ_{10} , its second crossing form is the nonzero scalar γ_{10} , and Proposition 5.3 gives

$$[x^{-2}] G_{10}(1/2, h_0 + x)^{-1} = \frac{\chi_{10} \chi_{10}^T}{\gamma_{10}}.$$

Taking the trace against the specialized torus matrix yields

$$[x^{-2}] F_{10}(1/2, h_0 + x, d) = \frac{\langle \chi_{10} | V_d(1) | \chi_{10} \rangle}{\gamma_{10}}.$$

For $N < 10$, the two singular submodules have zero intersection at grade N . Thus the radical dimension is

$$p(N - 2) + p(N - 4),$$

with $p(k) = 0$ for $k < 0$, and this equals the Kac-determinant valuation. Lemma 5.1 therefore shows that all lower-level Smith exponents at h_0 are at most one. Multiplication by the Euler product consequently does not alter the x^{-2} coefficient at grade 10. Finally, Lemma 5.5, applied along either side of the embedding diamond, gives

$$\langle \chi_{10} | V_d(1) | \chi_{10} \rangle = P_2(d) P_8^{(33/16)}(d) = P_4(d) P_6^{(65/16)}(d).$$

The Gram matrices and the normalized singular-vector coefficients are rational at $c = 1/2$, so the defining linear system gives $\gamma_{10} \in \mathbb{Q}$. It remains to evaluate this intrinsic scalar. Apply the generic recursion of Definition 6.1 algebraically near b_0 . The nested path $(1, 2) \rightarrow (4, 2)$ contributes

$$\frac{\dot{h}_{1,2} - \dot{h}_{2,2}}{\dot{h}_{1,2} - \dot{h}_{4,2}} \mathcal{A}_{1,2}(b_0) \mathcal{A}_{4,2}(b_0) P_2(d) P_8^{(33/16)}(d)$$

to the double pole. Lemma 6.3 evaluates its scalar coefficient as $-192/1121229484375$. The other path gives the same result. Comparison with the already proved intrinsic coefficient $P_2 P_8 / \gamma_{10}$ yields the displayed value of γ_{10} . $\square$

Theorem 6.5 (Complete grade-ten principal part). *Define the finite shifted-block parts*

$$Q_8(d) = \frac{\partial}{\partial z} \left\{ \left(z - \frac{33}{16} \right) H_8(1/2, z, d) \right\} \Big|_{z=33/16}$$

and

$$Q_6(d) = \frac{\partial}{\partial z} \left\{ \left(z - \frac{65}{16} \right) H_6(1/2, z, d) \right\} \Big|_{z=65/16}.$$

Put

$$\begin{aligned} \Theta_{10}(d) = & - \frac{d^2(d-1)^2(2d-7)^2(2d-1)^2(3d-14)(3d-5)(6d-55)(6d-1)}{552735279561465000000} \\ & \times (424640464d^8 - 41506307104d^7 + 1664502727528d^6 \\ & - 35552058801520d^5 + 438148123547677d^4 - 3142854705332170d^3 \\ & + 12488401942129875d^2 - 23870935242348750d + 14566042731240000). \end{aligned}$$

Then the entire principal part of the level-ten reduced coefficient at $\Delta = h_0 = 1/16$ is

$$\begin{aligned} \text{PP}_{\Delta=h_0} H_{10}(1/2, \Delta, d) = & - \frac{192 P_2(d) P_8^{(33/16)}(d)}{1121229484375 (\Delta - h_0)^2} \\ & + \frac{-\frac{4}{7} P_2(d) Q_8(d) + \frac{648}{13475} P_4(d) Q_6(d) + \Theta_{10}(d)}{\Delta - h_0}. \end{aligned}$$

Every term on the right is finite and is evaluated at $c = 1/2$.

Proof. Let $t = b - b_0$, and abbreviate the four labels

$$\alpha = (1, 2), \quad \beta = (2, 2), \quad \mu = (4, 2), \quad \nu = (1, 6).$$

Set

$$z_\alpha = h_\alpha + 2, \quad z_\beta = h_\beta + 4, \quad \delta_\alpha = z_\alpha - h_\mu, \quad \delta_\beta = z_\beta - h_\nu.$$

Only the α - and β -terms of the generic recursion can develop a pole at $\Delta = h_0$. Their numerators at grade ten are

$$C_\alpha(b) = \mathcal{R}_\alpha(b, d) H_8(c(b), z_\alpha(b), d), \quad C_\beta(b) = \mathcal{R}_\beta(b, d) H_6(c(b), z_\beta(b), d).$$

At $(b_0, z_\alpha(b_0))$, the level-eight shifted block has one transverse pole, labelled by μ , and no other Kac divisor through that point. The analogous statement holds for the level-six block and ν . Therefore the definitions of Q_8, Q_6 give

$$\begin{aligned} H_8(c(b), z_\alpha(b), d) &= \frac{\mathcal{R}_\mu(b, d)}{\delta_\alpha(b)} + Q_8(d) + O(t), \\ H_6(c(b), z_\beta(b), d) &= \frac{\mathcal{R}_\nu(b, d)}{\delta_\beta(b)} + Q_6(d) + O(t). \end{aligned}$$

For either nested pair (κ, τ) , expansion of $\mathcal{R}_\kappa \mathcal{R}_\tau / \delta_\kappa$ gives finite term

$$T_{\kappa, \tau} = \frac{\dot{\mathcal{R}}_\kappa \mathcal{R}_\tau + \mathcal{R}_\kappa \dot{\mathcal{R}}_\tau}{\dot{\delta}_\kappa} - \frac{\mathcal{R}_\kappa \mathcal{R}_\tau \ddot{\delta}_\kappa}{2\dot{\delta}_\kappa^2},$$

where every undotted factor on the right is evaluated at b_0 . The two t^{-1} terms cancel because

$$\frac{\mathcal{R}_\alpha \mathcal{R}_\mu}{\dot{\delta}_\alpha} + \frac{\mathcal{R}_\beta \mathcal{R}_\nu}{\dot{\delta}_\beta} = 0;$$

this is Lemma 6.3 together with $P_2P_8 = P_4P_6$.

Write $y = \Delta - h_0$. Expanding

$$\frac{C_\alpha(b)}{\Delta - h_\alpha(b)} + \frac{C_\beta(b)}{\Delta - h_\beta(b)}$$

at $t = 0$, the cancelled t^{-1} numerator produces the y^{-2} coefficient

$$\frac{\dot{h}_\alpha - \dot{h}_\beta}{\dot{\delta}_\alpha} \mathcal{R}_\alpha(b_0, d) \mathcal{R}_\mu(b_0, d) = -\frac{192}{1121229484375} P_2(d) P_8^{(33/16)}(d).$$

The y^{-1} coefficient is

$$\mathcal{R}_\alpha(b_0, d) Q_8(d) + \mathcal{R}_\beta(b_0, d) Q_6(d) + T_{\alpha, \mu} + T_{\beta, \nu}.$$

The first two residue constants are $-4P_2/7$ and $648P_4/13475$. Differentiating the finite products in Definition 6.1 and factoring over $\mathbb{Q}[d]$ gives

$$T_{\alpha, \mu} + T_{\beta, \nu} = \Theta_{10}(d).$$

Substitution proves the stated principal part. $\square$

Remark 6.6 (Exact replay). The PBW commutation, Ward recursion, determinant, rank, and null-vector checks in Propositions 5.7 and 5.8, as well as every residue identity in Theorem 5.9, are replayed by `tools/audit_low_levels.py`. The computation uses exact rational arithmetic. The Kac slopes, four generic residue constants, fusion factorization, value of γ_{10} , and polynomial Θ_{10} are replayed by `tools/audit_grade10_confluence.py`. The shifted Laurent recursion and its first two collision moments are independently replayed by `tools/audit_confluent_recursion.py`.

7 All-order confluent recursion

Definition 7.1 (Shifted Laurent recursion). Let $t = b - b_0$, and work in

$$K((t)), \quad K = \mathbb{Q}(i\sqrt{3}, d).$$

For a positive Kac label $a = (r, s)$, write

$$\ell_a = rs, \quad h_a(t) = h_{r,s}(b_0 + t), \quad R_a(t) = \mathcal{R}_{r,s}(b_0 + t, d), \quad z_a(t) = h_a(t) + \ell_a.$$

Define shifted Laurent coefficients recursively by

$$S_{a,0}(t) = 1$$

and, for $M \geq 1$,

$$S_{a,M}(t) = \sum_{\substack{e=(u,v) \in \mathbb{Z}_{\geq 0}^2 \\ \ell_e = uv \leq M}} \frac{R_e(t)}{z_a(t) - h_e(t)} S_{e, M - \ell_e}(t).$$

Every denominator in this formula is a nonzero rational function of b , and the recursion terminates because its second index strictly decreases.

Lemma 7.2 (No generic shifted self-poles). *For positive labels $a = (r, s)$ and $e = (u, v)$,*

$$h_{r,s}(b) + rs - h_{u,v}(b)$$

is not the zero rational function of b .

Proof. The Kac formula gives the identity

$$h_{r,s}(b) + rs = h_{r,-s}(b).$$

If this were identically equal to $h_{u,v}(b)$, then

$$(rb - sb^{-1})^2 = (ub + vb^{-1})^2$$

in $\mathbb{C}(b)$. Since this is an integral domain,

$$rb - sb^{-1} = \pm(ub + vb^{-1}).$$

Comparison of the coefficients of b and b^{-1} contradicts $r, s, u, v > 0$. Thus every shifted denominator in Definition 7.1 has a well-defined Laurent expansion. $\square$

Definition 7.3 (Collision moments). For $N \geq 1$, set

$$\mathcal{C}_{n,N} = \{a = (r, s) \in \mathbb{Z}_{>0}^2 : \ell_a \leq N, |4r - 3s| = n\}.$$

For $a \in \mathcal{C}_{n,N}$, define

$$B_{a,N}(t) = R_a(t)S_{a,N-\ell_a}(t), \quad \delta_a(t) = h_a(t) - \lambda_n.$$

If $\delta_a \neq 0$ and $B_{a,N} \neq 0$, put

$$e_a = \text{ord}_t \delta_a, \quad v_{a,N} = \max\{0, -\text{ord}_t B_{a,N}\}.$$

If $\delta_a \neq 0$ and $B_{a,N} \equiv 0$, put $e_a = \text{ord}_t \delta_a$ and $v_{a,N} = 0$. If $\delta_a \equiv 0$, set $e_a = \infty$ and $v_{a,N} = 0$. Define the finite structural pole bound

$$M_{n,N} = \max_{a \in \mathcal{C}_{n,N}} \begin{cases} 1 + \lfloor v_{a,N}/e_a \rfloor, & e_a < \infty, \\ 1, & e_a = \infty, \end{cases}$$

and, for $1 \leq k \leq M_{n,N}$, define

$$A_{n,k,N}(d) = \text{CT}_t \sum_{a \in \mathcal{C}_{n,N}} B_{a,N}(t) \delta_a(t)^{k-1}.$$

Here CT_t denotes the coefficient of t^0 in a Laurent series. Empty collision classes contribute zero. The actual pole multiplicity at grade N is

$$m_{n,N} = \max(\{k : A_{n,k,N} \neq 0\} \cup \{0\}),$$

and satisfies $m_{n,N} \leq M_{n,N}$.

Theorem 7.4 (Pole-free Ising recursion and all-order equality). *For every $N \geq 1$,*

$$H_N(1/2, \Delta, d) = \sum_{\substack{n \geq 0 \\ \mathcal{C}_{n,N} \neq \emptyset}} \sum_{k=1}^{M_{n,N}} \frac{A_{n,k,N}(d)}{(\Delta - \lambda_n)^k}.$$

The sums and every constant-term calculation are finite, $A_{n,k,N}(d) \in \mathbb{Q}[d]$, and the recursion in Definitions 7.1–7.3 terminates after grades strictly below N . Consequently, with

$$A_{n,k}(d; q) = \sum_{N \geq 1} A_{n,k,N}(d) q^N,$$

one has the coefficientwise locally finite identity

$$H_{\Delta}^{(1/2)}(d \mid q) = 1 + \sum_{n \geq 0} \sum_{k \geq 1} \frac{A_{n,k}(d; q)}{(\Delta - \lambda_n)^k}.$$

This series agrees coefficient by coefficient with Definition 2.2, hence with the inverse-Shapovalov definition of the torus one-point block.

Proof. For generic b , the eta-reduced recursion in Definition 6.1 gives

$$H_N(c(b), \Delta, d) = \sum_{\ell_a \leq N} \frac{R_a(t)}{\Delta - h_a(t)} H_{N-\ell_a}(c(b), z_a(t), d).$$

Induction on M in the same recursion gives

$$S_{a,M}(t) = H_M(c(b), z_a(t), d)$$

for generic b . Thus

$$H_N(c(b), \Delta, d) = \sum_{\ell_a \leq N} \frac{B_{a,N}(t)}{\Delta - h_a(t)}. \quad (6.1)$$

All sums are finite.

Fix N , and choose Δ outside the finite set of Ising Kac weights occurring at grade N . By Lemma 2.4, the inverse-Shapovalov expression is analytic in t near zero: its specialized Gram determinant is nonzero at this Δ . Equation (6.1), initially valid for generic b , is an identity of rational functions and therefore holds on the punctured formal neighborhood of b_0 .

Group its right-hand side according to the distinct limits λ_n :

$$\mathcal{G}_{n,N}(t, \Delta) = \sum_{a \in \mathcal{C}_{n,N}} \frac{B_{a,N}(t)}{\Delta - h_a(t)}.$$

Each coefficient of a negative power of t in $\mathcal{G}_{n,N}$ is a rational function of Δ , tends to zero as $\Delta \rightarrow \infty$, and has its poles only at $\Delta = \lambda_n$. The sum of these negative Laurent coefficients over n vanishes because the left-hand side of (6.1) is analytic in t . Partial-fraction uniqueness for the distinct λ_n 's forces the negative coefficient of each $\mathcal{G}_{n,N}$ to vanish separately. Hence every collision class has a finite algebraic confluent value.

Put $x = \Delta - \lambda_n$. In the formal expansion at $x = \infty$,

$$\frac{1}{\Delta - h_a(t)} = \frac{1}{x - \delta_a(t)} = \sum_{j \geq 0} \frac{\delta_a(t)^j}{x^{j+1}}.$$

Taking CT_t after summing the collision class shows that the coefficient of x^{-k} is exactly

$$\text{CT}_t \sum_{a \in \mathcal{C}_{n,N}} B_{a,N}(t) \delta_a(t)^{k-1} = A_{n,k,N}(d).$$

If $\delta_a \neq 0$, then the summand has positive t -valuation whenever $k > 1 + \lfloor v_{a,N}/e_a \rfloor$, so its constant term vanishes. If $\delta_a \equiv 0$, it vanishes identically for $k > 1$. This proves the bound $M_{n,N}$, the finiteness of the pole sum, and the finite-jet nature of every constant-term calculation. The confluent value of each class tends to zero as $\Delta \rightarrow \infty$ and has no finite pole except λ_n ; therefore these finitely many moments are its complete partial-fraction expansion, with no polynomial remainder.

Every $R_a(t)$ is polynomial in d with coefficients rational in t , and all shifted denominators are independent of d . Therefore every collision moment is polynomial in d . Equation (6.1) and the analyticity argument identify the resulting rational function with the specialized inverse-Shapovalov coefficient. The latter has rational coefficients at $c = 1/2$; uniqueness of its partial fractions improves $A_{n,k,N} \in K[d]$ to $A_{n,k,N} \in \mathbb{Q}[d]$. Finally, local finiteness in q is Corollary 3.3, completing the all-order identity. $\square$

Corollary 7.5 (Recovery of the first two confluent formulas). *At $N = 4$, Theorem 7.4 gives Theorem 5.9. At $N = 10$, its $\lambda_2 = 1/16$ collision moments give Theorem 6.5.*

Proof. At $N = 4$, the class $\mathcal{C}_{2,4}$ consists of $(1, 2)$ and $(2, 2)$, and both B -terms are regular at $t = 0$. The first collision moment is

$$-\frac{4}{7}P_2(d)H_2(1/2, 33/16, d) + \frac{648}{13475}P_4(d),$$

while all higher moments vanish.

At $N = 10$, the same first-label class has

$$B_{(1,2),10} = \mathcal{R}_{1,2}H_8(c(b), h_{1,2} + 2, d), \quad B_{(2,2),10} = \mathcal{R}_{2,2}H_6(c(b), h_{2,2} + 4, d).$$

Their zeroth and first collision moments are exactly the two Laurent coefficients computed in the proof of Theorem 6.5; all higher moments vanish by the grade-ten Smith profile. $\square$

8 Irreducible Ising specialization

Proposition 8.1 (Identity insertion in the Verma recursion). *For $d = 0$,*

$$H_{\Delta}^{(1/2)}(0 \mid q) = 1$$

as a rational function of Δ . Thus specialization of the meromorphic Verma block at a degenerate weight is not the irreducible Ising trace; the latter requires taking the quotient by the radical.

Proof. In Definition 6.1, the factor with $k = l = 1$ in $\mathcal{P}_{r,s}(b, 0)$ is

$$\frac{(b + b^{-1})^2 - (b + b^{-1})^2}{4} = 0.$$

Hence every generic residue $R_a(t)$ vanishes identically at $d = 0$. Definitions 7.1 and 7.3 then give $A_{n,k,N}(0) = 0$ for every $N \geq 1$. Equivalently, in the inverse-Shapovalov definition the identity insertion has $\rho_N = G_N$, so $F_N = p(N)$ and eta reduction gives $H = 1$. $\square$

Definition 8.2 (Eta-reduced irreducible identity block). For $h \in \{0, 1/2, 1/16\}$, let $L(1/2, h)$ be the irreducible quotient of $M(1/2, h)$, and define

$$\widehat{H}_h(q) = \prod_{m \geq 1} (1 - q^m) \sum_{N \geq 0} \dim L(1/2, h)_{h+N} q^N.$$

Theorem 8.3 (The three Ising characters). *The quotient identity blocks are*

$$\begin{aligned} \widehat{H}_0(q) &= \sum_{j \in \mathbb{Z}} \left(q^{((24j+1)^2-1)/48} - q^{((24j+7)^2-1)/48} \right), \\ \widehat{H}_{1/2}(q) &= \sum_{j \in \mathbb{Z}} \left(q^{((24j+5)^2-25)/48} - q^{((24j+11)^2-25)/48} \right), \end{aligned}$$

and

$$\widehat{H}_{1/16}(q) = \sum_{j \in \mathbb{Z}} \left(q^{((24j-2)^2-4)/48} - q^{((24j+10)^2-4)/48} \right).$$

Consequently,

$$\chi_h(q) = q^{h-1/48} \frac{\widehat{H}_h(q)}{\prod_{m \geq 1} (1 - q^m)}$$

are the vacuum, energy, and spin Ising characters. Their eta-reduced beginnings are

$$\begin{aligned} \widehat{H}_0 &= 1 - q - q^6 + q^{11} + q^{13} - q^{20} + \dots, \\ \widehat{H}_{1/2} &= 1 - q^2 - q^3 + q^7 + q^{17} - q^{25} - q^{28} + \dots, \\ \widehat{H}_{1/16} &= 1 - q^2 - q^4 + q^{10} + q^{14} - q^{24} - q^{30} + \dots. \end{aligned}$$

Proof. The Ising model is the minimal series $c_{3,4} = 1/2$. The Class R^+ , Case 1+, BGG resolution in [1, Theorem 6.9] is an exact complex of Verma modules. Taking its Euler–Poincaré character gives, for an interior label (r, s) ,

$$\chi_{r,s}(q) = \frac{1}{\eta(q)} \sum_{j \in \mathbb{Z}} \left(q^{(24j+4r-3s)^2/48} - q^{(24j+4r+3s)^2/48} \right).$$

Use $(r, s) = (1, 1), (2, 1), (1, 2)$, respectively. Since

$$h_{r,s} = \frac{(4r-3s)^2 - 1}{48}, \quad \eta(q) = q^{1/24} \prod_{m \geq 1} (1 - q^m),$$

factoring out $q^{h_{r,s}-1/48}$ gives exactly the three displayed $\hat{H}$ -series. For any fixed power of q , only finitely many integers j contribute, so the Euler–Poincaré sums are coefficientwise finite. Direct evaluation of the exponents through degree 30 gives the final three lines. $\square$

Remark 8.4 (Character replay). The three BGG numerators and the resulting quotient dimensions through grade 15 are replayed by `tools/audit_ising_characters.py` using exact integer arithmetic.

References

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